

Q.1)

House	X	Y	
H ₁	2	2	-
H ₂	2	4	-
H ₃	5	8	-
H ₄	6	7	-
H ₅	9	2	-
H ₆	10	3	-

$C^1: T_1 = H_1(2,2)$
 $C^2: T_2 = H_3(5,8)$

initial

$K = 2$ Euclidean distance

Step 1

House	X	Y	T ₁	T ₂	Tower
H ₁	2	2	0	5	T ₁
H ₂	2	4	2	0	T ₁
H ₃	5	8	0	0	T ₂
H ₄	6	7	6.40	1.41	T ₂
H ₅	9	2	7	7.21	T ₁
H ₆	10	3	8.06	7.07	T ₂

ex 1

$$\sqrt{(6-2)^2 + (7-2)^2} = \sqrt{41}$$

$$\sqrt{(6-5)^2 + (7-8)^2} = \sqrt{2}$$

T₁ : H₁, H₂, H₅

T₂ : H₃, H₄, H₆

C₁¹ : (2,2), (2,4), (9,2)

C₁² : (5,8), (6,7), (10,3)

(centroid for cluster 1)
 $\left(\frac{2+2+9}{3}, \frac{2+4+2}{3} \right)$

$\left(\frac{5+6+10}{3}, \frac{8+7+3}{3} \right)$

C₁¹ : (4.33, 2.67)

C₁² : (7, 6)

✓

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	Hour	X	Y	C_1^0	C_1^1	T_2
T_1	H_1	2	2	-		
T_1	H_2	2	4	-		
T_2	H_3	5	8	-		
T_2	H_4	6	7	-		
T_2	H_5	9	2	-		
T_2	H_6	10	3	-		

H_1, H_2

H_3, H_4, H_5, H_6

$C_2^0 : (2, 3)$

$C_2^1 : (7.5, 5)$

$C_3^0 : (2, 3)$

$C_3^1 : (7.5, 5)$

Step 3:

Q. 2)

	Point	<u>Nbr (dist ≤ 0.6)</u>	Nbr (count)	Type
			$\{P_1, P_8, P_{11}\}$	3 = B
-	$P_1 (1, 2)$		$\{P_2, P_3, P_6\}$	3 = B
	$P_2 (3, 4)$		$\{P_2, P_3, P_6, P_7\}$	4 = C
→	$P_3 (2.5, 4)$		$\{P_4, P_8, P_{11}\}$	3 = B
	$P_4 (1.5, 2.5)$		$\{P_5, P_6\}$	2 = B
	$P_5 (3, 5)$		$\{P_2, P_3, P_5, P_6\}$	5 = C
-	$P_6 (2.8, 4.5)$		$\{P_3, P_6, P_7\}$	3 = B
	$P_7 (2.5, 4.5)$		$\{P_1, P_4, P_8, P_9, P_{11}\}$	5 = C
	$P_8 (1.2, 2.1)$		$\{P_8, P_9, P_{11}\}$	3 = B
→	$P_9 (1, 3)$		$\{P_{10}\}$	1 = <u>Nain</u>
	$P_{10} (1, 5)$		$\{P_1, P_4, P_8, P_9\}$	4 = C
	$P_{11} (1, 2.5)$			

Core, Border, Nain

C	B	N
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$\sqrt{(1-1.5)}$

$C1: \{P_1, P_3, P_6, P_7\}$
 $\{P_2, P_3, P_5, P_6, P_7\}$
 $\{P_2, P_3, P_5, P_6, P_7\}$
 $C2: \{P_1, P_4, P_8, P_9, P_{11}\}$
 $\{P_1, P_4, P_8, P_{11}\}$

Nain
 P_{10}

Q.)
①

S.L.
= men C_i, C_j

	C_1 18	C_2 22	C_3 25	C_4 27	C_5 42	C_6 43
C_1 18	0	4	7	9	24	25
C_2 22	4	0	3	5	20	21
C_3 25	7	3	0	2	17	18
C_4 27	9	5	2	0	15	16
C_5 42	24	20	17	15	0	1
C_6 43	25	21	18	16	1	0

$\sqrt{16}$
↓
4

②

		18	22	25	27	(42, 43)
C_1	18	0	4	7	9	24
C_2	22	4	0	3	5	20 ←
C_3	25	7	3	0	2	17
C_4	27	9	5	2	0	15
C_5	(42, 43)					0

→ ③

		18	22	(25, 27)	(42, 43)
C_1	18	0	4	7	24
C_2	22	4	0	3	20
C_3	(25, 27)				
C_4	(42, 43)				

⑤

